BM5033 - Statistical Inference Methods in Bioengineering (Stage - II)

BM23BTECH11015	AI22BTECH11008	EE22BTECH11214	BM25MTECH14005
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(a) The Experimental design remains unchanged provided the dataset.

- We compare CBF/MFV means of Diabetic and Control Group

H_0(Null hypothesis) : $(\mu_{diabetes})_{TCD} = (\mu_{non\text{-}Diabetic})_{TCD}$
H_a(Alternate hypothesis) : $(\mu_{diabetes})_{TCD} < (\mu_{non\text{-}Diabetic})_{TCD}$

H_0(Null hypothesis) : $(\mu_{diabetes})_{MRI} = (\mu_{non\text{-}Diabetic})_{MRI}$
H_a(Alternate hypothesis) : $(\mu_{diabetes})_{MRI} < (\mu_{non\text{-}Diabetic})_{MRI}$

(b) The sample size we got is 13 based on our power analysis for CBF measured using MRI. also the sample size of 18 for MFV measured using TCD. But since we have more than 18 samples in both Diabetic and non-Diabetic we have randomly chosen 18 & 13 samples from each data set using a python script.

```
power.t.test(n = NULL, delta = 5.0, sd = 5.906, sig.level = 0.05,
             power = 0.8,
             type = c("two.sample"),
             alternative = c("one.sided"))
```

```
     Two-sample t test power calculation

              n = 17.96882
          delta = 5
             sd = 5.906
      sig.level = 0.05
          power = 0.8
    alternative = one.sided

NOTE: n is number in *each* group
```

```
power.t.test(n = NULL, delta = 0.05, sd = 0.049, sig.level = 0.05,
             power = 0.8,
             type = c("two.sample"),
             alternative = c("one.sided"))
```

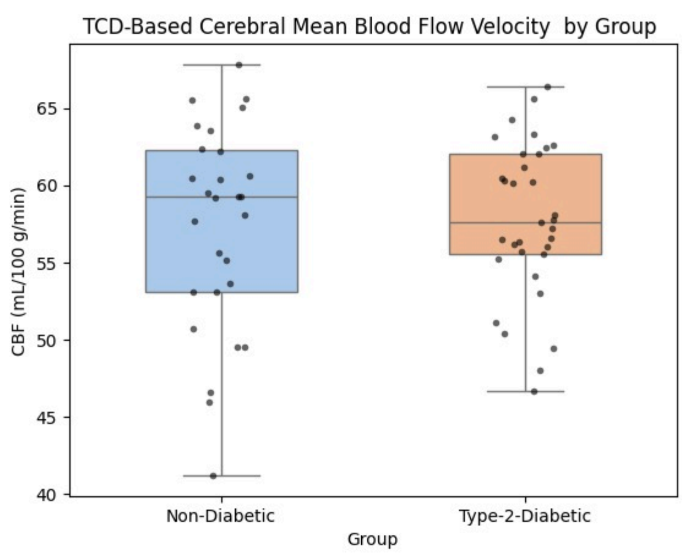
```
     Two-sample t test power calculation

              n = 12.61046
          delta = 0.05
             sd = 0.049
      sig.level = 0.05
          power = 0.8
    alternative = one.sided

NOTE: n is number in *each* group
```

(c) Graphically represent the data in a concise manner. It should highlight the question you are trying to answer.

Data Description:

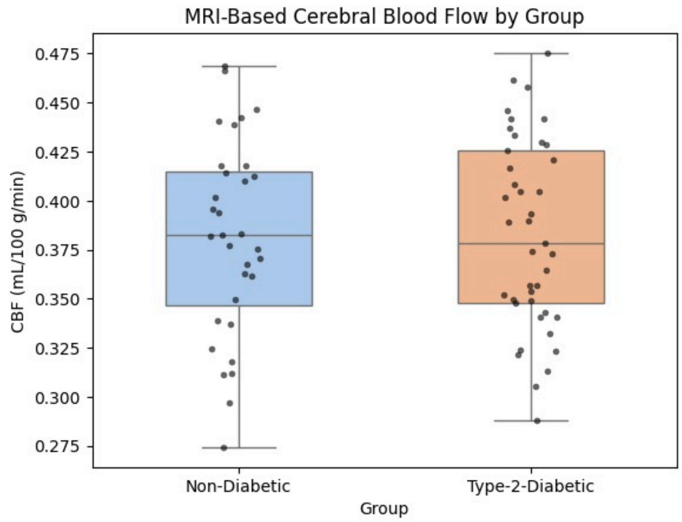
- MRI data were taken from GE-79_Summary_Table-Demographics-MRI-Part1.csv. Consider Global GM as CBF of whole brain
- TCD features (MFV, PI) were extracted from raw Doppler recordings(in labview2) from the middle cerebral artery (MCA). instead of preprocessed files.
- When two visits existed, only one i.e visit 2 was used to keep the study cross-sectional.

Box Plot between Cerebral Blood flow velocity between non-diabetic and type-2 diabetic using TCD



- **Median CBF:** The median cerebral blood flow velocity(MFV) in Type-2 Diabetic participants is slightly lower than in Non-Diabetic participants.
- **Variability:** The Non-Diabetic group shows slightly greater variability in MFV values compared to the Type-2 Diabetic group.
- **Overlap:** There is considerable overlap between the two groups, suggesting no large difference in overall MFV distribution.
- **Preliminary inference:** Visual inspection suggests that Type-2 Diabetes may be associated with marginally reduced cerebral blood flow.

Box Plot between Cerebral Blood flow between non-diabetic and type-2 diabetic using MRI



- **Median CBF:** The median cerebral blood flow (CBF) in Type-2 Diabetic participants is similar in Non-Diabetic participants.
- **Variability:** The Non-Diabetic group shows slightly greater variability in CBF values compared to the Type-2 Diabetic group.
- **Overlap:** There is considerable overlap between the two groups, suggesting no large difference in overall CBF distribution.
- **Preliminary inference:** Visual inspection suggests that Type-2 Diabetes has almost the same cerebral blood flow.

(d) Analysing the data using Statistical Test (one tailed T-test)

After selecting the 18 and 13 samples from each of Diabetic and Non-diabetic and taking the MFV(which can be measured by TCD) and CBF(can be measured by MRI) respectively we then calculated the means and tested for the t-test to each of them indicidually R code for the same is as follows: (code used for the selecting 13 and 18 random samples from the given CSV file)

```
# MFV comparison
res_MFV = t.test(x = DMFV, y = NDMFV,
       alternative = "less",
       paired = FALSE, var.equal = FALSE,
       conf.level = 0.95)
print("MFV comparison result:")
show(res_MFV)

# CBF comparison
res_CBF = t.test(x = DCBF, y = NDCBF,
       alternative = "less",
       paired = FALSE, var.equal = FALSE,
       conf.level = 0.95)
print("CBF comparison result:")
show(res_CBF)
```

```
import pandas as pd

# Load dataset
df = pd.read_csv("df_merged.csv")

# Randomly select 20 Non-Diabetic and 20 Type-2-Diabetic rows
non_diabetic_sample = df[df['Group'] == 'Non-Diabetic'].sample(n=18, random_state=42)
type2_diabetic_sample = df[df['Group'] == 'Type-2-Diabetic'].sample(n=18, random_state

# Extract CBF and MFV values
non_diabetic_cbf = non_diabetic_sample['CBF'].tolist()
non_diabetic_mfv = non_diabetic_sample['MFV'].tolist()
type2_diabetic_cbf = type2_diabetic_sample['CBF'].tolist()
type2_diabetic_mfv = type2_diabetic_sample['MFV'].tolist()

# Print results clearly
print("Non-Diabetic CBF:")
print(", ".join(map(str, non_diabetic_cbf)))

print("\nNon-Diabetic MFV:")
print(", ".join(map(str, non_diabetic_mfv)))

print("\nType-2-Diabetic CBF:")
print(", ".join(map(str, type2_diabetic_cbf)))

print("\nType-2-Diabetic MFV:")
print(", ".join(map(str, type2_diabetic_mfv)))
```

Now let's see the results of both t-tests.

```
[1] "MFV comparison result:"

    Welch Two Sample t-test

data:  DMFV and NDMFV
t = 0.67977, df = 25.832, p-value = 0.7487
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
     -Inf 4.625391
sample estimates:
mean of x mean of y
 57.96778  56.64990

[1] "CBF comparison result:"

    Welch Two Sample t-test

data:  DCBF and NDCBF
t = 0.78683, df = 23.964, p-value = 0.7805
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
      -Inf 0.0449229
sample estimates:
mean of x mean of y
0.4049136 0.3907626
```

We can clearly see that p value in both cases is not less than 0.05 which is our minimum value to reject the null hypothesis. So we fail to reject the null hypothesis in both the cases.

Justification:

CBF/MFV values for both groups overlap heavily (nearly same mean and spread) for given data, the t-test naturally fails to detect a difference even if one exists physiologically.

(e) Summarize your findings and conclusions

- The study examined whether Type-2 Diabetes Mellitus (T2DM) affects cerebral microcirculation using MRI and TCD data from the CDed dataset.
- **MRI results:** No significant differences in global or regional gray-matter perfusion between diabetic and non-diabetic groups. Large-scale brain perfusion appeared similar.
- **TCD results:** Small but not so significant differences in mean flow velocity, indicating changes in blood flow in diabetic participants.
- **Conclusion:** TCD is a little more sensitive for detecting early or mild vascular changes in diabetes compared to MRI which does not show a significant difference in CBF.